

Lens-free on-chip vibrometer with millimetre working distance

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Abstract *On-chip vibration sensors usually rely on bulk optics to deliver the sensing beam to the moving target, thereby limiting integration. We show that large optical antennas enable target distances of ~5mm while achieving measurement resolutions of 12nm, without using any bulk optics. ©2026 The Authors*

Introduction

Interferometric optical sensors offer unparalleled spatial resolutions compared to electro-mechanical solutions [1], and their integration into optical chips is of great interest to achieve rugged, miniaturized solutions [2].

For such integrated systems, several approaches have been explored. Optical feedback interferometry exploits variations in laser cavity power when light reflected from the target is fed back into the cavity. On-chip solutions based on this technique have reported resolutions of 5.6nm when a collimator was used to direct light towards the target, which however degraded to 50nm without the collimator [3]. In [4], directional emission from a Huygens dipole was exploited to achieve a positioning accuracy of 6nm but required a microscope objective with a 0.9 numerical aperture for light coupling. A system based on an on-chip free propagation region reported a position noise of the order of a few nanometers for a target at a 1cm distance, to which light was coupled using a collimating lens [5]. Homodyne detection uses an optical hybrid to recombine a reference signal with the signal reflected back from the target thereby recovering their relative phase and enabling unambiguous tracking of the movement. Such systems have achieved noise equivalent displacements of the order of 6nm, using bulk optical lenses to couple light from the chip to the target and back [6]. Heterodyne systems achieve picometre resolutions but require a significantly more complex chip with integrated modulators and still rely on specialized external probe-heads for focusing and collecting light from the target [7]. Finally, micro-electromechanical systems (MEMS) based sensors have reported picometre resolutions [8], [9]. However, this approach requires the sensor to be in direct contact with the moving target, since the MEMS is directly integrated with the chip.

Here, we present an on-chip homodyne sensor [see Fig. 1(a)] that can measure nanometre displacements of targets situated at centimetre-scale distances, without requiring any bulk optics elements, thereby significantly facilitating integration. This is achieved by using a large optical antenna [see Fig. 1(b)] that emits a highly collimated

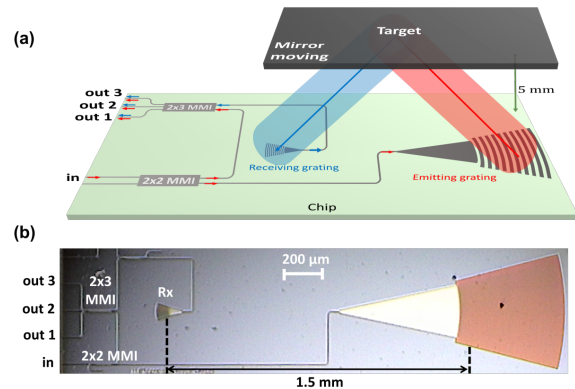


Fig. 1: (a) Schematic of the on chip vibrometer including the emitting and receiving optical antennas and the beam splitting and combining multimode interference couplers (not to scale). (b) Optical microscope image of the vibrometer on a silicon on insulator chip (Tx/Rx: transmitting/receiving antenna).

beam which barely diverges as it propagates towards the target. The chip is fabricated on standard 220nm SOI with a single etch step. To the best of our knowledge, our experimental measurement resolution of 12nm at a 5mm distance is not only competitive compared to the state-of-the-art, but also the best figure reported so far for lens-free non-contact sensors.

Operation and design

The vibrometer chip is shown schematically in Fig. 1(a) – it is essentially a Mach-Zehnder interferometer, in which one arm consists of free-space propagation to the target (a moving mirror) and back. The basic operation of the device is as follows. Light is coupled in and out from the left side of the chip through a fibre array. Light injected in the “in” port, is split by a 50/50 coupler (implemented by a 2×2 multimode interference coupler – MMI), and is directed either to the emitting antenna, or to the first input of the output 2×3 MMI where it serves as a reference signal. The emitting antenna radiates a large optical beam that barely diffracts as it propagates towards the mirror and reflects towards the receiving antenna. The receiving antenna is significantly smaller, to reduce its angular selectivity and thus enhance alignment tolerances. The signal collected by the

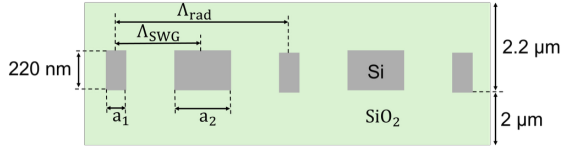


Fig. 2: Side view of the grating geometry.

receiving antenna is then routed to the second input of the 2×3 MMI. This device produces three interferometric output signals [labelled out 1, 2 & 3 in Fig. 1(a)], from which the relative phase between the reference signal and the signal reflected by the mirror can be recovered without directional ambiguity or loss of sensitivity [10], [11], thereby overcoming the limitations of classical solutions [12]. This relative phase is directly proportional to the displacement of the target. Fig. 1(b) shows the implementation of the vibrometer on a 220nm silicon-on-insulator chip with a single, full etch step, a $2\mu\text{m}$ bottom-oxide layer and a $2.2\mu\text{m}$ silicon dioxide top cladding. The chip was fabricated using e-beam lithography followed by reactive ion etching at Applied Nanotools [13].

We now briefly describe the design of the key elements of the device. Coupling between the chip and the fibre-array is optimized by using sub-wavelength patterned inverse tapers that enable virtually polarization independent coupling [14]. Both the emitting and receiving optical antennas operate with TM polarization at a wavelength of 1550nm and are based on the fully etched dual-period subwavelength structure described in [15]. This structure was combined with in-plane focusing to achieve an emitting antenna with a compact layout and a mode field diameter of $200\mu\text{m}$; the corresponding Rayleigh distance of 20.3mm ensures that the radiated beam barely diverges as it propagates. The receiving antenna is designed to exhibit a significantly smaller mode field diameter of $20\mu\text{m}$. The size difference between the emitted beam and the receiving grating results in coupling efficiency penalty of 15dB. However, it also relaxes the angular alignment tolerance due to the broader radiation pattern of the receiving antenna. This enables light collection with $\pm 0.6^\circ$ angular tolerance in both the roll and pitch axes of the mirror, at the cost of an additional 5.5dB penalty. The dimensions of the gratings are illustrated in Fig. 2, where $\Lambda_{\text{rad}} = 2\Lambda_{\text{SWG}}$. For the emitting antenna, $\Lambda_{\text{rad}} = 898\text{nm}$, $a_1 = 111\text{nm}$ and $a_2 = 277\text{nm}$. Similarly, for the receiving antenna $\Lambda_{\text{rad}} = 854\text{nm}$, $a_1 = 113\text{nm}$ and $a_2 = 385\text{nm}$. The on-chip distance of 1.5mm between the gratings [see Fig. 1(b)] is determined trigonometrically by the target distance (5mm in this case) and the radiation angle of the gratings (our designs radiate “backwards” at an angle of -8.5 degrees with respect to the chip normal). We

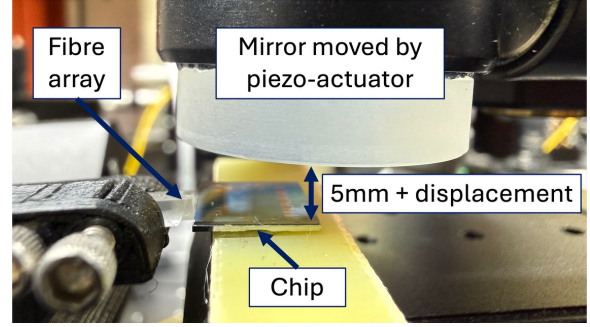


Fig. 3: Part of the measurement setup used to characterize the vibrometer chip.

note that using large mode-field-area gratings that radiate perfectly vertical, as reported in [16], would enable operation at a wide range of target distances. The 2×2 and 2×3 MMIs are optimized for TM polarization and use tapered $1.5\mu\text{m}$ wide input and output waveguides. The dimensions of the multimode region are $5\mu\text{m} \times 69\mu\text{m}$ for the 2×2 device and $7.45\mu\text{m} \times 99\mu\text{m}$ for the 2×3 device. The interconnecting waveguides are single-mode ($0.5\mu\text{m}$ wide).

Measurement setup

To experimentally characterize the device, we used the setup that is shown partially in Fig. 3. Light from a fixed laser source at 1550nm (SANTEC, WSL-100) with an output power of 7dBm is sent through a fibre polarization controller (Thorlabs, FPC562) and coupled into the “in” port of the chip with a fibre array (MEISU, PN: 30112126) mounted on a six-axis alignment stage (Thorlabs Nanomax 600/M). A second three-axis stage (Thorlabs, PT3A/M) with a two-axis angular gimbal (Thorlabs, GMB1/M) is used to align a dielectric mirror (Thorlabs, BB111-E04) approximately 5mm above the chip. The mirror can then be vertically displaced using a multi-axis nanopositioning stage (Physik Instrumente, P-762) driven by a piezoelectric actuator (Physik Instrumente, E-501.00). The actuator exhibits a linear relation between input voltage and displacement of $\alpha = 10\mu\text{m}/\text{V}$. The three output signals [out 1, 2 & 3 in Fig. 1(a)] are collected with the fibre array and are simultaneously detected with three amplified photodetectors (Thorlabs, PDA20CS2) with their transimpedance gain adjusted to 15kV/A (resulting in a bandwidth of 1MHz) to match the dynamic range of the data acquisition card. The three resulting electrical signals are digitized with a 16bit data acquisition board (National Instruments, NI USB-6356) at a sampling rate of 100kHz and stored on a computer for processing. The speed of the displacement that can be detected is in principle only limited by the detection electronics, not the chip itself.

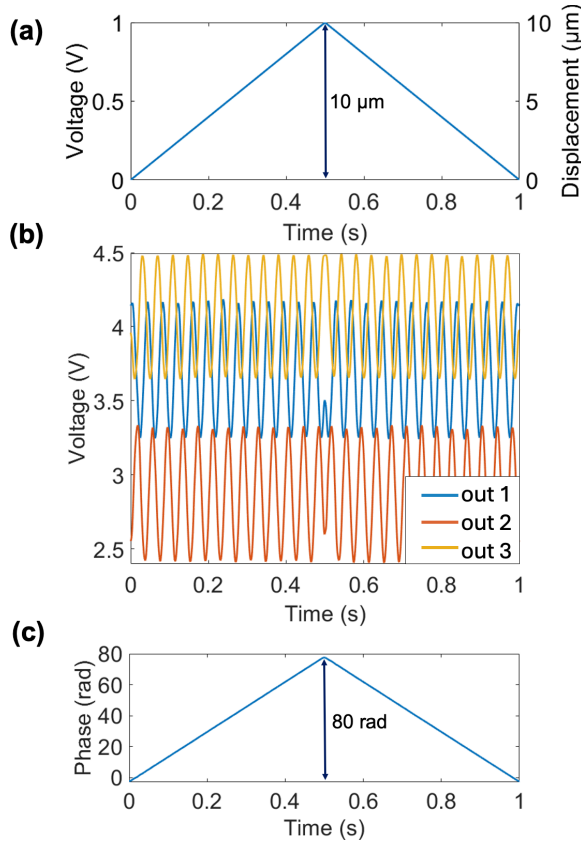


Fig. 4: (a) Driving voltage of the piezo-electric actuator and nominal displacement of the mirror. (b) Measured interfero-

Results and discussion

The measurement process is illustrated in Fig. 4 (a)-(c). We drive the piezo-electric actuator that moves the mirror vertically with a triangular wave at 1Hz between 0V and 1V resulting in a nominal displacement between 0μm and 10μm, as shown in Fig. 4(a). This results in the sinusoidal output voltages shown in Fig. 4(b), which are proportional to $\cos((2\pi/\lambda_0)2d(t)/\cos(8.5^\circ) + \phi_i)$, where $d(t)$ is the vertical mirror displacement, $\cos(8.5^\circ)$ accounts for the -8.5° emission angle of the antenna, $\lambda_0=1550\text{nm}$, and $\phi_{\{1,2,3\}}=\{-2\pi/3, 0, 2\pi/3\}$ are the relative phase shifts imparted by the 2×3 MMI. The three signals in Fig. 4(b) should ideally have the same mean and peak-to-peak values, however, due to uneven coupling to the fibre array they are offset. This is however easily corrected via blind calibration and after linearly combining the three output signals [10], the complex signal $\exp(j(2\pi/\lambda_0)2d(t)/\cos(8.5^\circ))$ is obtained. The phase of this signal [Fig. 4(c)] is directly proportional to the displacement. Indeed, the nominal displacement of 10μm shown in Fig. 4(a) results a phase shift of 80rad, which corresponds to the expected sensitivity of $4\pi/(\lambda_0 \cos 8.5^\circ) = 8.2 \text{ rad}/\mu\text{m}$.

To test the resolution of the vibrometer we measured the displacement when the mirror was still, as shown in Fig. 5, and obtain a noise deviation just below 4nm [see inset Fig. 5], which,

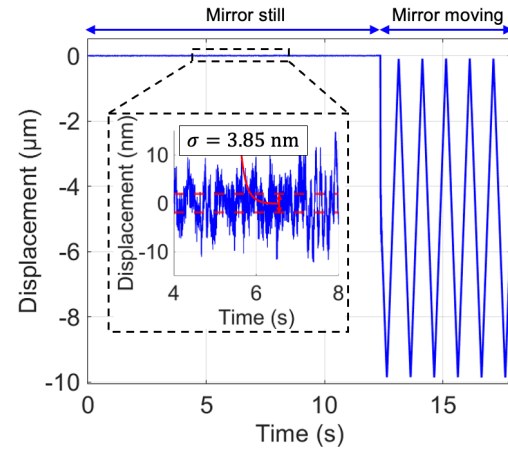


Fig. 5: Measured displacement when the mirror is still and when piezo-electric actuator is driven by a triangular voltage.

using the 3-sigma criterion indicates a resolution just below 12nm. The main limiting factor is the small spurious movement of the head of the fibre array with respect to the chip, which introduces a small modulation of the output signals. This could be overcome by glueing the array head to the chip. Still, as shown in Table 1, our resolution is comparable to that of state-of-the art solutions most of which use bulk optics to direct light towards the target. Compared with the only other lens-free solution [3], our resolution is more than four times better.

Tab. 1: Comparison of integrated vibrometer solutions.

Ref.	Year	Technique	Lens-free	Res. [nm]
[3]	2024	Feedback interferometry	No	5.6
[3]	2024	Feedback interferometry	Yes	50
[4]	2020	Directional emission	No	6
[5]	2016	Young interferometer	No	~3
[6]	2013	Homodyne detection	No	6
[7]	2023	Heterodyne detection	No	0.01
This work	2026	Homodyne detection	Yes	12

Conclusions

We have experimentally demonstrated an on chip vibrometer that does not require any bulk optics to couple light to the target. The achieved resolution is on par with the state of the art of much more complex devices that use bulk optics elements and provides a four-fold improvement compared to the other lens-free integrated solution.

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